7 Conclusion

We asked how geometry, supports and loads shape the deformation of a two-dimensional elastic object. A hole makes the plate softer, and holes and inner corners concentrate stress; moving the load on the bracket moves the stress peak. Holding fewer displacement components barely changes the displacement but raises $\kappa_2(A)$. Slender bending is the hardest case: linear triangles are too stiff in bending, and even the finest beam mesh deflects 0.7% less than beam theory with shear predicts.

Since A is sparse, symmetric and positive definite, sparse direct factorization and CG both apply, and the CG bound predicts iterations growing like $\sqrt{\kappa_2(A)}$. We measured $\kappa_2(A)$ growing like N and CG iterations like $N^{1/2}$ for every object; shape changes the constant, and the beam is by far the worst conditioned. CG is the best iterative method. Diagonal preconditioning saves few iterations and no time, Jacobi diverges and Gauss–Seidel is impractically slow. On the beam MINRES stalls above the residual target although its solution is accurate. At these sizes sparse LU is the fastest method on every system.

Under refinement the displacements settle, but peak stresses do not: at the inner corner the peak keeps growing (119 to 178 MPa), as the infinite exact corner stress predicts. The algebraic error of the accepted CG solutions, at most 10^{-9}, is far below the mesh changes, so the mesh limits the accuracy. The solutions satisfy the supports exactly and balance force and moment, a load perturbation stays within the condition-number bound, and removing the vertical support makes A numerically singular, as expected. The main limitations, detailed in Section 6, are that only diagonal preconditioning was tested, constant element stresses make peaks mesh dependent, and the linear model ignores yielding. Natural next steps are incomplete Cholesky or multigrid preconditioners, larger or three-dimensional problems in which iterative methods may overtake sparse LU, and better stress recovery or higher-order elements.